Math 133 – Business Calculus – Fall 2026

Course Overview and Basic Information

Welcome to Math 133! My goal is to introduce you to the fundamentals of differential calculus, financial mathematics, and applications of these topics to business.

The key to success in our course will be to spend enough time outside of class (both individually and with your classmates) engaging with course material. Successful students will likely spend **at least six hours outside of class** each week on the course. This includes reviewing course notes, discussing ideas with classmates, and working on homework assignments. The key to **learning** mathematics is **doing** mathematics — the most effective way to learn this material is to work on problems.

Most weeks, we will have online homework, one written homework assignment, and a quiz. There will be two midterms and one final exam.

All course meetings will take place in **Saints Hall 314** on **Mondays, Wednesdays, and Fridays from 11:15 a.m.–12:10 p.m.**.
Instructor: Christian Klevdal
Email: cklevdal@sandiego.edu
Office: Saints Hall 146
Office hours: Mondays from 9:00–10:00 am, Wednesdays from 1:30–2:30 pm, and Fridays from 9:00–10:00 am in Saints Hall 146.

> Office hours are for your benefit. Please feel free to stop by my office at any point during office hours. You don't need to have a specific question — you can just drop in to see what's going on.

Please stay home if you are sick. Your health is more important than attending a class meeting. If you need to miss class due to an illness, let me know as soon as you are able, and we will work together to get you caught up on course material.

Course website. We will use Canvas for course announcements and materials.

I am committed to your success. Please feel free to email me at any time. A collection of USD resources is included at the end of this document.

> DISCLAIMER: Math 133 is **not** equivalent to Math 150 (Calculus I), and it is **not** a prerequisite for Math 150 or Math 151. If you have any plans that may include taking Math 150 or 151, then Math 133 is likely **not** the correct course for your needs.

Mathematics Learning Center (MLC): The MLC is a dedicated mathematics tutoring center, located in Saints Hall 134. The MLC is a great place to work on Math 133 homework even when you don't have specific questions. Visit the MLC website for a current schedule of open hours.

Course Content

Text: We will use the **OpenStax Business Calculus textbook**:

https://www2.math.uconn.edu/ClassHomePages/Math1071/Textbook2/OpenStaxBusinessCalc.html

We will use the following supplementary texts and resources:

- Mike May and Anneke Bart, *Business Calculus with Excel*. This free text emphasizes business modeling, spreadsheet work, marginal analysis, and optimization.
- OpenStax, *College Algebra 2e*. We will use selected sections for algebra review and possible concluding topics.

Prerequisites: Math 115 or departmental placement test.

Learning outcomes: Math 133 introduces differential calculus through mathematical modeling and applications to business. By the end of the course, you will

- interpret functions and mathematical models presented symbolically, graphically, numerically, and verbally, including appropriate domains, ranges, and units;
- construct and analyze business models involving supply and demand, revenue, cost, profit, growth, and marginal quantities;
- calculate and interpret average rates of change, limits, continuity, and derivatives of algebraic, exponential, and logarithmic functions;
- use derivatives to estimate change, analyze function behavior, and solve constrained optimization problems, and interpret the results in context;
- clearly communicate and evaluate mathematical reasoning using correct notation, terminology, and supporting evidence.

This course satisfies the **Mathematical Reasoning and Problem Solving (CMRP)** competencies area of the core curriculum. For this, students will demonstrate:

1. *Mathematical problem solving:* Apply mathematical methods to solve problems including problems with applications to other disciplines.
2. *Mathematical reasoning, argumentation and proof:* Demonstrate mathematical reasoning by being able to a) create chains of mathematical arguments, including using definitions and theorems appropriately, and b) assess chains of mathematical arguments.
3. *Mathematical explanation:* Clearly communicate mathematical reasoning and solutions to problems by using correct mathematical notation, terminology and symbolism.

Topics and Course schedule

Topics: Functions and mathematical modeling, including algebraic, graphical, numerical, and spreadsheet representations; linear, quadratic, rational, piecewise, exponential, and logarithmic business models; average rates of change, limits, continuity, derivatives, and differentiation rules; and marginal analysis, linear approximation, curve analysis, elasticity of demand, and constrained optimization. As time permits, we will conclude with either geometric sums and financial mathematics or matrices and systems of linear equations.

Schedule: This a tentative schedule of the intended content and pacing. This is only meant to serve as a guide; the timing will change in response to the needs of the class.

Week	**Dates**	**Topics and important dates**
1	Sept. 1–4	Functions, domain and range, mathematical models, and spreadsheet graphing. Tuesday, Sept. 1 follows the Monday class schedule.
2	Sept. 7–11	Graph behavior, quadratic extrema, and elementary constrained optimization. No class Monday, Sept. 7.
3	Sept. 14–18	Business Modeling I: linear, quadratic, and nonlinear models.
4	Sept. 21–25	Business Modeling II: exponential and logarithmic models.
5	Sept. 28–Oct. 2	Average rates of change and marginal change; review; Midterm 1 on Friday.
6	Oct. 5–9	Limits and continuity: from average to instantaneous change.
7	Oct. 12–16	The derivative and elementary differentiation rules.
8	Oct. 19–23	Product, quotient, and chain rules.
9	Oct. 26–30	Exponential and logarithmic derivatives and marginal analysis.
10	Nov. 2–6	Linear approximation; review; Midterm 2 on Friday.
11	Nov. 9–13	Extrema, concavity, curve sketching, and asymptotic behavior.
12	Nov. 16–20	Analytical optimization with constraints.
13	Nov. 23–27	Spreadsheet optimization with constraints. Only Monday meets because of the Thanksgiving holiday.
14	Nov. 30–Dec. 4	Elasticity of demand and extended optimization.
15	Dec. 7–11	Optional concluding module: geometric sums and financial mathematics, or matrices and systems of linear equations.

Grades

Your grade in Math 133 will be determined by your performance in the following areas.

Participation	5%
Homework	25%
Quizzes	20%
Midterm 1	15%
Midterm 2	15%
Final	20%

Cutoffs for letter grades will be no higher than 90% for an A-, 80% for a B-, 70% for a C-, and 60% for a D-. I may decide to lower these when I determine overall course grades, but they are guaranteed not to increase. Here is more information about each of the above categories:

- **Participation:** Your participation grade is based on the following, all of which will be graded based on completion.

  - Completing the Entry Survey by Sunday, September 6.
  - Attendance and active participation during class meetings. Simply attending class is not enough: I expect you to work on problems and discuss them with classmates when prompted, and you should not spend significant class time on non-course material. You can miss four class meetings without penalty, no questions asked. Please talk to me if you need to miss more.
  - Our classroom is a **no-phone zone**. If I notice that you are regularly using your phone during class, I will take points from your participation grade. Please talk to me if you have a legitimate need to use a phone during class.

  Additional participation requirements may be added.

- **Homework:** We will have both online homework and written homework (submitted on Gradescope).

  - **Online Homework:** 10% of grade. Online homework will be due every Tuesday at 11:59 p.m. from Week 2 through Week 15. Late work will not be accepted, and no online homework assignments will be dropped. You will have unlimited attempts on each problem. The purpose of online homework is to provide additional practice. I suggest first attempting each problem on your own, without using the textbook, AI, internet searches, or help from friends, so that you can accurately gauge your understanding. If you get stuck, use an appropriate resource to help, then reflect afterward on what you understand and what still needs work. The online homework platform will be announced on Canvas.
  - **Written Homework:** 15% of grade. There will be one written homework assignment due most weeks, starting in Week 2. Each assignment is due at 11:59 p.m. on the Wednesday shown on the course website. Late work will not be accepted. These assignments will be submitted on Gradescope and graded largely based on thoughtful completion, so it is important that you use them to learn rather than simply maximize your score on each problem. Submissions must be legible. Clearly identify each problem, show enough reasoning that another student in the course could follow your work, and assign the correct page or pages to each problem when uploading your submission. The main purpose of homework is to practice course concepts and prepare for quizzes and exams. You should attempt each problem yourself and write up your own solution, although your work may draw on discussions with classmates. I will drop the two lowest written homework scores before calculating your course grade.

    Use of AI: You must disclose any use of generative AI, including the tool used and how you used it. Do not use AI as your first approach to a problem. First make a serious attempt on your own and consult course resources such as the textbook and lecture notes. AI may support, but may not replace, your own reasoning; anything you submit must be written in your own words and must be work that you understand and can explain. Copying or lightly editing an AI-generated solution, or failing to disclose AI use, is a violation of the course's academic integrity policy.

- **Quizzes:** Starting in week 2, we will have a quiz at the beginning of class each Friday that will focus on material covered earlier in the course. These are meant to be low-pressure and are primarily for you and me to determine whether you are on track with course material. I will drop the two lowest quiz scores before calculating your course grade.

- **Exams:** There will be two midterm exams and a final exam. If your final exam score is higher than your lowest midterm score, your final exam score will replace your lowest midterm score when calculating your course grade. More specifics and test policies will be posted later. The exams are scheduled as follows. The final exam date and time are fixed and cannot be changed.

Midterm 1	Friday, October 2, during class
Midterm 2	Friday, November 6, during class
Final	Friday, December 18, 11:00 a.m.–1:00 p.m.

You are welcome to use a simple scientific calculator (e.g., a TI-30XIIS) on quizzes and exams, but no questions on these assessments will require the use of a calculator. More advanced calculators (e.g., graphing calculators or ones with significant amounts of storage) are not allowed; please contact me if you have questions about a particular calculator.

Please be aware that any grade or score calculated by the Canvas Gradebook (e.g., combined homework or quiz averages) may not accurately reflect your actual Math 133 grade.

USD Resources

The University of San Diego has many resources that are free to you, including:

- Student Wellness Center, which aims to provide a "comprehensive and integrated range of wellness programs, experiences and services" to students.
- Counseling Center, whose purpose is "to enhance the emotional, relational, and psychological well-being of students." For 24/7 access to a counselor, call **(619) 260-4655** and press 1 for urgent concerns.
- Disability and Learning Difference Resource Center, which is "committed to helping students with disabilities obtain meaningful academic accommodations and support."
- C.A.R.E. Advocates are USD's "primary effort to provide support, resources and education to the student community pertaining to sexual assault and relationship violence." You can call a CARE Advocate anytime at **(619) 260-2222**.
- Student Health Center's mission is to "provide primary healthcare and to promote the health and wellbeing of students."
- Onward USD is the central hub for COVID-19 information on campus.
- USD Food Pantry provides "food, school supplies, hygiene supplies, laundry detergent" and other items. Every USD student is welcome to use the pantry.

Academic Integrity

I take academic integrity very seriously. Please review the university's Academic Integrity Policy. Be aware that working with your classmates on assignments is **not** academic misconduct — in fact I **strongly encourage** you to work with other Math 133 students outside of exams and quizzes. However, using non-approved sources (e.g., other textbooks, the internet, AI, etc.) without proper attribution is academic misconduct and is subject to penalties.